

FRANKIE OKOTH

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SUMMARY

Systems Software Engineer specializing in high-concurrency backend infrastructure and deterministic multi-agent AI architectures. Focused on engineering low-latency data pipelines and sovereign, reproducible computing environments.

TECHNICAL SKILLS

Languages:	Python · C++ · JavaScript · TypeScript · SQL · Bash
Frameworks & Libraries:	Django · REST APIs · Node.js · React.js (Three.js, Framer Motion) · Tailwind CSS
Systems & Infrastructure:	Linux (Ubuntu/WSL2) · Docker · Git · CI/CD Pipelines · Makefiles · GNU Stow · Google Cloud
Databases:	PostgreSQL · MongoDB

WORK EXPERIENCE

Telkom Kenya (Nairobi, Kenya) - Software Engineering Intern *July - Sep 2023*

- Developed T-Kash mobile-money features by building Django microservices and optimizing PostgreSQL schemas to process real-time financial data flows for a nationwide platform.
- Rewrote core backend validation logic to resolve transactional workflow defects and mitigate system failure rates across high-availability APIs.
- Executed internal vulnerability scanning and security assessments to harden transaction processing modules and enforce strict authorization protocols.
- Configured Docker environments on Linux to containerize backend modules and streamline production deployment pipelines.

PROJECTS

Sauti Ya Mwananchi: Multi-agent civic intelligence stack (Node.js, TypeScript, Gemini 2.5 Flash) featuring a 3-layer deterministic guardrail gate and zero-retention ephemeral memory; secured 3rd place at the Build with AI Agentathon 2026 (GDG Nairobi).

Lemmon-714: Modular, reproducible development environment (engineered via GNU Stow, Bash, Makefiles) running on Ubuntu 24.04 LTS (WSL2) to eliminate host-machine configuration drift using symlink-based automation.

EDUCATION

University of Eastern Africa, Baraton (Eldoret, Kenya) *Aug 2020 - Aug 2025*

- Bachelor of Science in Software Engineering
- Upper Second-Class Honours
- Selected Coursework: Data Structures & Algorithms · Operating Systems · Database Systems · Software Architecture & Design · Cloud Computing · Object-Oriented Programming

CERTIFICATIONS & ACHIEVEMENTS

AI Fundamentals - IBM	<i>June 2024</i>
Cloud Skills Challenge Badge (Microsoft Build) - Microsoft	<i>June 2024</i>